

E-COMMERCE PRODUCT CATALOGUE

1. **Abstract:** Online shopping has become an everyday activity for millions of users, yet many e-commerce platforms struggle to provide a smooth and intuitive browsing experience. This project, the E-Commerce Product Catalogue, aims to solve that problem by building a feature-rich, responsive web application that makes product discovery easy and enjoyable. The application supports multi-level product categories, allowing users to drill down from broad categories to specific subcategories. A powerful filter panel lets shoppers narrow results by price, brand, rating, colour, and size simultaneously. An intelligent search bar offers real-time suggestions as the user types, significantly reducing the time taken to find a desired item. Users can also compare multiple products side by side, helping them make more informed purchase decisions. The solution is built using React.js on the frontend for a dynamic, component-driven interface, with Node.js and Express.js powering the backend REST APIs. State management is handled through Context API or Redux, and the UI is styled using Bootstrap or Tailwind CSS for a clean, professional look. An optional MySQL database layer supports persistent product and user data storage. The project consists of over ten distinct modules including a product grid, category navigation, filter panel, search bar, sort options, compare tool, product detail view, wishlist, browsing history, and an admin panel for content management. Together, these modules deliver a complete, scalable, and maintainable e-commerce catalogue experience.

MODULE 1 – Product Grid

Purpose

Displays all products in a clean grid layout.

Features

- Product image
- Product name
- Price

- Rating
- Add to wishlist button

Technologies Used

- React Components
- CSS Grid / Flexbox
- Axios API calls

Working

- Fetch products from backend
- Store data in state
- Render products dynamically

MODULE 2 – Category Navigation

Purpose

Helps users browse products category-wise.

Features

- Multi-level categories
- Dropdown navigation
- Sidebar categories

Example

Electronics → Mobiles → Android Phones

Technologies Used

- React Router
 - Recursive Components
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MODULE 3 – Filter Panel

Purpose

Allows users to filter products.

Filters Included

- Price
- Brand
- Rating
- Colour
- Size

Working

When user selects filters:

- Filter state updates
- Product list updates dynamically

Technologies Used

- Context API / Redux
 - Controlled Components
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MODULE 4 – Search Bar

Purpose

Helps users quickly find products.

Features

- Real-time suggestions
- Keyword matching
- Instant results

Technologies Used

- Axios
- Debouncing technique
- React Hooks

Working

As user types:

- API request is sent
 - Matching products are shown
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MODULE 5 – Sort Options

Purpose

Sorts products based on user preference.

Sorting Methods

- Price Low to High
- Price High to Low
- Rating
- Newest Products

Working

Sorting function rearranges products dynamically.

MODULE 6 – Product Detail View

Purpose

Displays complete information about a product.

Information Displayed

- Product image
- Description
- Price
- Ratings
- Available sizes
- Reviews

Technologies Used

- React Router parameters
 - Dynamic API fetching
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MODULE 7 – Compare Tool

Purpose

Allows users to compare multiple products.

Features

- Side-by-side comparison
- Compare specifications
- Compare prices

Example

Compare:

- Mobile A vs Mobile B

Working

Selected products are stored temporarily and displayed in table format.

MODULE 8 – Wishlist

Purpose

Allows users to save favorite products.

Features

- Add to wishlist
- Remove from wishlist
- Persistent storage

Technologies Used

- LocalStorage
 - React State
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MODULE 9 – Browsing History

Purpose

Tracks recently viewed products.

Features

- Recently viewed items
- Quick revisit option

Working

Product IDs are stored in LocalStorage.

MODULE 10 – Admin Panel

Purpose

Allows admin to manage products.

Features

- Add products

- Edit products
- Delete products
- Manage categories

Technologies Used

- Forms
 - CRUD operations
 - Express APIs
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